MATLB cv. 6 - Analýza funkcí

Funkce:

$$f(x) = tan^{-1}\frac{x+1}{x-1} \tag{1}$$

$$g(x) = \frac{x}{\sqrt{x^2-1}} \tag{2}$$

$$h(x) = x^x \tag{3}$$

Definiční obory:

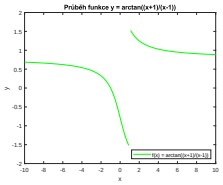
- $Df(x) = \mathbb{R} \setminus \{1\}$
- $Dg(x) = (-\infty; -1) \cup (1; \infty)$
- $Dh(x) = (0; \infty), x \in \mathbb{R}$

Limity:

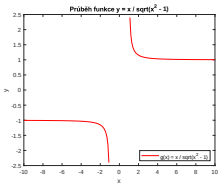
Tabulka 1: Hodnoty funkcí v limitních bodech

Limita	$f(x)$	$g(x)$	$h(x)$
$\lim_{x\to\infty}$	$\frac{\pi}{4}$	1	∞
$\lim_{x\to-\infty}$	$\frac{\pi}{4}$	-1	0
$\lim_{x\to1^+}$	$\frac{\pi}{2}$	∞	1
$\lim_{x\to1^-}$	$\frac{-\pi}{2}$	$\sqrt{-\infty}$	1

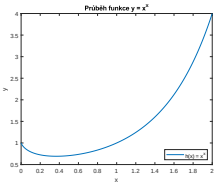
Grafy funkcí:



(a) $f(x)$



(b) $g(x)$



(c) $h(x)$

Obrázek 1: Průběh analyzovaných funkcí